Structural Drivers of the U.S. Overdose Crisis:

Evidence from State-Level Social Determinants

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Content Overview

01 Introduction

02 Data

03 Methodology

04 Empirical Results

05 Conclusion and Policy Implications

Introduction-Overdose Crisis in The U.S.

Not just about drugs, but the social roots behind

Surged Overdose Mortality

~91,799 overdose deaths in 2020

~31% increase from 2019

Social Determinants of Health (SDOH)

Multiple SDOH Domains

Insurance coverage
- Barriers to treatment access

Food insecurity
- Chronic economic stress exposure

Mental health
- Depression elevates overdose risk

Sleep health
- Sleep disruption fuels substance misuse

Disability
- Chronic pain and vulnerability

Introduction-Research Questions & Hypotheses

Research Questions

Primary question

- Are state-level overdose death rates associated with socioeconomic conditions, mental health, and physical health factors?

Secondary question

- Do these associations change over time as the overdose crisis evolves?

Hypotheses

H1. Economic pressure

- Higher uninsurance and food insecurity → higher overdose mortality

H2. Mental & physical health burden

- Higher depression, short sleep, and disability → higher overdose mortality

H3. Temporal dynamics

- The strength of these associations may vary over time

Introduction - Data, Methods & Findings

Data

- CDC WONDER:
- CDC PLACES
- 51 states + DC, 2011-2020

Methods

- State-year panel
- OLS with linear time trend
- **2 Models**

Key Findings

- **Positive associations**: depression, short sleep, uninsurance
- Food insecurity: weak and unstable
- **Negative associations:** disability (contrasts with micro evidence)
- **Insurance effect weakens over time**

Data

Drug Overdose Mortality Data

CDC WONDER

- Number of deaths
- Crude death rates per 100,000 residents
- Age-adjusted death rates

Social Determinants of Health Data

CDC PLACES

- Share of adults uninsured
- Food insecurity
- Depression

Methodology-Variable Definition

Response Variable

- **Morality of Overdose**
  Crude_Rate

Explanatory Variable

- **Economic Pressure**
  Food_Insecurity
  Lack_Insurance

- ***Individual Health Status***
  Depression
  Short sleep
  Disability

Methodology-Model Setup

Model I

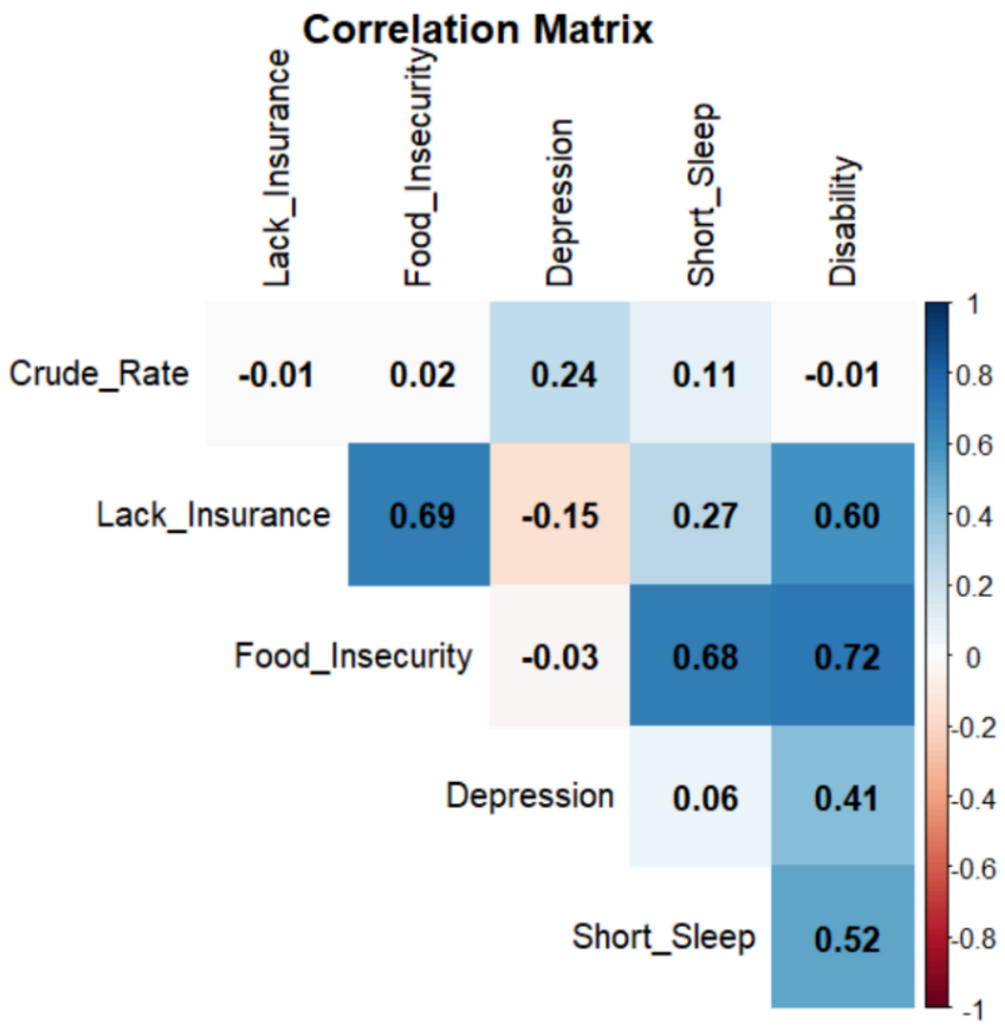
$$\text{Crude_Rate} = \beta 0 + \beta 1 * \text{Year} + \beta 2 * \text{Lack_Insurance} + \beta 3 * \text{Food_Insecurity} + \beta 4 * \text{Depression} + \beta 5 * \text{Short sleep} + \beta 6 * \text{Disability} + \varepsilon$$

Model II

$$\text{Crude_Rate} = \beta 0 + \beta 1 * \text{Year} * \text{Lack_Insurance} + \beta 2 * \text{Year} * \text{Food_Insecurity} + \beta 3 * \text{Depression} + \beta 4 * \text{Short sleep} + \beta 5 * \text{Disability} + \varepsilon$$

Empirical Results-Descriptive Statistics

	mean	min	max
Crude_rate(%)	15.16	2.00	78.30
Lack_Insurance(%)	10.40	5.15	20.70
Food_Insecurity(%)	13.78	10.28	26.42
Depression(%)	23.29	16.18	31.42
Short sleep(%)	36.05	30.44	43.02
Disability(%)	32.12	21.80	40.58



Empirical Results- Regression outcome

Model I

	Estimate	P-Value
Year	0.9267	***
Lack_Insurance	1.0020	***
Food_Insecurity	0.1778	
Depression	1.4549	***
Short_Sleep	0.7416	***
Disability	-1.0694	***

Fail to reject H1,H2

Model II

	Estimate	P-Value
Year	1.5503	***
Depression	1.4586	***
Short_Sleep	0.7435	***
Disability	-1.0727	***
Year:Lack_Insurance	-0.0801	***
Year:Food_Insecurity	0.0109	

Fail to reject H3

Empirical Results- Diagnostics and Comparison

Model Comparison(ANOVA):Explanatory power was enhanced

Residual sum of squares (RSS) :

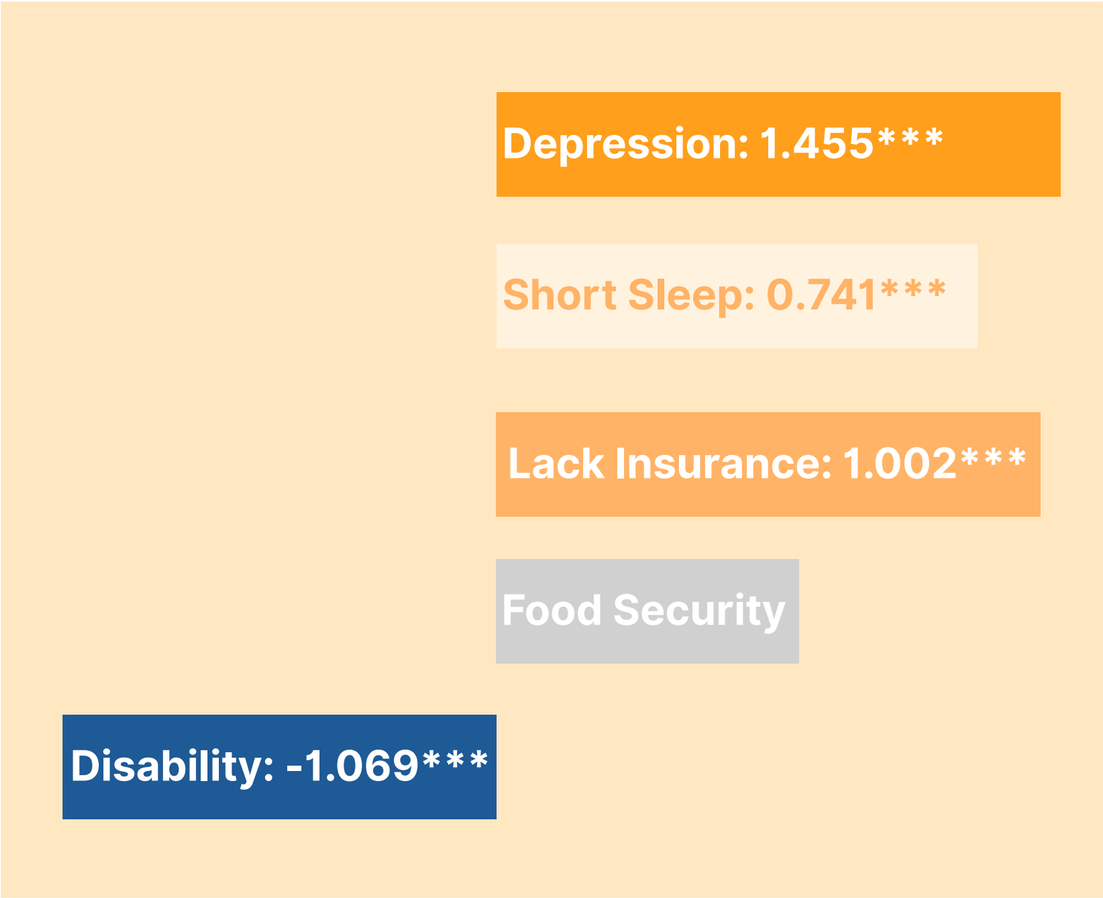
Model I: 28132

Model II: 27020

Model Diagnostics(VIF): There's no severe multicollinearity issues

Year	Lack_Insurance	Food_Insecurity	Depression	Short_Sleep	Disability
1.0002	2.8277	4.5203	1.9927	2.2492	4.2742

Conclusion



Mental and behavioral health are key drivers

- The crisis is a symptom of broader psychological distress and lifestyle deterioration.

The protective role of health insurance has weakened over time

- no longer a "silver bullet."
- Low-quality insurance is not enough to protect vulnerable populations.

State-level patterns do not equal individual risks

- We cannot simply assume that trends seen in whole states apply directly to individuals.

Policy Implications

Integrate Mental Health and Sleep Hygiene

Overdose prevention funds should be allocated to community mental health services and public health campaigns promoting sleep hygiene.

Focus on Quality, Not Just Coverage

Policy should shift focus from merely expanding coverage to improving the quality of coverage.

Targeted Interventions

Interventions must be continuous and focus on regions with high combined risks of mental distress and economic instability.

THANK YOU